MIL-R-52411B
18 December 1984
SUPERSEDING
MIL-R-52411A
12 July 1978

MILITARY SPECIFICATION

RAMS, HYDRAULIC, 10,000 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single-acting push and push-pull hydraulic rams.

1.2 Classification. Hydraulic rams shall be of the following types and capacities, as specified (see 6.2):

Type I - Single-acting, push ram.

Capacity - 4-Ton.
Capacity - 7-Ton.
Capacity - 10-Ton.
Capacity - 20-Ton.
Capacity - 50-Ton.

Type II - Single-acting, push-pull ram.

Capacity - 17-1/2-Ton.
Capacity - 20-Ton.
Capacity - 30-Ton.
Capacity - 50-Ton.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The hydraulic ram shall consist of cylinder(s), piston(s), piston plunger control valve, and all accessories necessary to form an operable ram.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.2). Any changes or deviations of rams from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish rams conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Materials deterioration and control. The hydraulic ram shall be fabricated from capatible materials, inherently corrosion resistant or treated to be resistant to the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by a barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components or sub-components, and shall make information available upon request to the contracting officer or designated representative.

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3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the rams may be newly fabricated from recovered materials to the maximum extent practicable, provided the rams produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the rams.

3.4 Screw-thread. Screw-threads shall be in accordance with FED-STD-H28.

3.5 Ram. The ram shall be single- or double-cylinder, single-acting and shall operate at not more than 10,000 pounds per square inch (psi) (69 Mega Pascals [MPa]) working pressure when using hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083.

3.5.1 Pressure loss. After the hydraulic pump and hose assembly have been disconnected from the control valve, the ram shall support the load without a loss of working pressure of more than 500 psi (3.5 MPa) for the first 4 hours. For the next 4 hours, the pressure loss shall not exceed 20 psi (0.14 MPa) when tested as specified in 4.5.2.3.

3.5.2 Hydrostatic. The ram shall withstand an internal pressure of not less than 12,000 psi (82.7 MPa) for a period of not less than 5 minutes without failure, leakage, or permanent deformation of any component.

3.6 Control valve. The ram shall be equipped with a manual control, positive shut-off needle valve, and a nipple. The valve shall have 0.375-inch NPT internal threads on both ends. The valve and nipple shall operate at a working pressure of not less than 10,000 psi (69 MPa). The nipple shall have 0.375 inch NPT external threads on both ends. The valve and nipple shall withstand a pressure of not less than 12,000 psi (82.7 MPa) for a period of not less than 5 minutes without failure, leakage, or permanent deformation of any component.

3.7 Identification marking. The ram shall be identified in accordance with MIL-STD-130. The marking shall be applied to the ram on plates conforming to MIL-P-514, type I, style 3, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plates will be visible and legible.

3.8 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the rams normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.9 Type I - Single-acting, push ram. The type I ram shall be single acting and shall push the rated-capacity loads specified in table I. The cylinder, piston, and piston plunger shall be steel. Bearing surfaces shall be materials which provide minimum friction and long life and prevent galling of rubbing surfaces. While at 75 percent of the fully extended position and with the control valve closed, the ram shall support the rated-capacity of loads specified in table I without leakage or permanent deformation of any component.

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TABLE II. Single-acting, push-pull ram.

Rated capacity tons	Plunger dia. inches (mm)	Stroke inches (mm)	Collapsed height inches (mm)	Extended height inches (mm)	Internal pressure at capacity max psi (MPa)	Weight max pounds (Kg)
17-1/2	1.75 (44.5) max	2.00 (50.8) ±.0625 (±1.59)	7.00 (177.8) max	9.063 (230.2) max	10,000 (69)	13 (5.90)
20	2.00 (50.8) max	2.00 (50.8) ±.0625 (±1.59)	6.375 (161.9) max	8.438 (214.3) max	10,000 (69)	17 (7.71)
30	2.25 (57.2) max	2.50 (63.5) ±.0625 (±1.59)	8.50 (215.9) max	11.00 (279.4) max	10,000 (69)	27 (12.25)
50 max	3.50 (88.9) ±.0625	3.00 (76.2) max (±1.59)	10.75 (273.1) max	13.813 (350.8)	10,000 (69)	65 (29.48)

3.10.1 Accessories. The type II ram shall be furnished with a solid serrated saddle, adjusting screw, speed crank, and speed unit. The center hole of the ram shall have internal threads to receive the external threads of the adjusting screw. The speed nut shall have internal threads to mate with external threads of the adjusting screw. The speed nut shall have internal threads to mate with external threads of the adjusting screw. The speed crank shall rotate the adjusting screw without slippage, breakage, or permanent deformation to the speed crank, adjusting screw, or speed nut.

3.10.1.1 Threads. Threads for type II ram and its accessories shall be as specified in table III.

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TABLE III. Threads, type II ram.

Rated cap (tons)	Saddles (external) shank	Ram (center hole)	Adjusting screw (external)	Speed nut (internal)
17-1/2	1-8UNC-2A	1-8UNC-2B	1-8UNC-2A	1-8UNC-2B
20	1-8UNC-2A	1-8UNC-2B	1-8UNC-2A	1-8UNC-2B
30	1.25-7UNC-2A	1.25-7UNC-2B	1.25-7UNC-2A	1.25-7UNC-2B
50	1.625-5.50 UNC-2A	1.625-5.50 UNC-2B	1.625-5.50 UNC-2A	1.625-5.50 UNC-2B

3.11 Workmanship. All parts, components, and assemblies of the ram, including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered.

3.12 Human factors engineering. The 10,000 psi hydraulic rams shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, paragraphs 4 (General requirements), 5.4 (Controls), 5.5 (Labeling), 5.6 (Anthropometry), 5.9 (Design for maintainability), and 5.13 (Hazards and safety) of MIL-STD-1472, as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The ram shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The ram shall be tested as specified in 4.5.2.1 through 4.5.2.3; and in addition, the type II ram shall be tested as specified in 4.5.2.4. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 4.0 percent defective for major defects and 10.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each ram shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2 through 4.5.2.4, if applicable (see 4.3.2). AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The ram shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage operating environment.
103. Dissimilar metals as defined in MIL-STD-889 not insulated from one another.
104. Contractor does not have documentation available for identifiction of material, material finishes or treatments.
105. Components missing or not as specified.
106. Weights not as specified.
107. Dimensions not as specified.
108. Control valve not provided or not as specified.
109. Accessories not as specified.
110. Screw-threads not as specified.
111. Workmanship not as specified.

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117. Cushioning and blocking not provided for consolidated items as specified for level A.
118. Box closure not method V as specified for level A.
119. Shipping container not as specified for level A and B.
120. Strapping not as specified for level A and B.
121. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Ram. The interior surfaces of the ram shall be coated with hydraulic fluid approved for operation of the ram. Openings into the ram shall be sealed with caps or plugs or with tape conforming to PPP-T-60, type IV.

5.1.1.2 Technical publications. The publications for each ram shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.3 Repair parts. The preservative application criteria of MIL-P-116 shall be used to preserve the repair parts. Tools shall be preserved in accordance with PPP-P-40, level A.

5.1.1.4 Consolidation. Each ram, technical publications, and repair parts items shall be consolidated in a close-fitting box conforming to PPP-B-636, class weather resistant, style optional. Cushioning and blocking shall be provided to control movement of items within the container. Box closure shall be in accordance with the appendix to the box specification, method V.

5.1.2 Commercial. Each ram, technical publications and repair parts items shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Rams of like description preserved as specified in 5.1, shall be packed in quantities not to exceed 250 pounds in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2) strapping shall be finish B. When specified (see 6.2) strapping shall be finish A.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except that the boxes shall be domestic type and strapping shall be finish A.

5.2.3 Commercial. Rams preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The hydraulic rams are intended for general purpose applications where a push or pull is required. The rams may be used with a hydraulic pump conforming to MIL-P-52310.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and capacity of ram required (see 1.2).
c. Time frame required for submission of first produced ram, and number of rams required (see 3.2).
d. Treatment and painting required when other than that specified (see 3.8).
e. Degree of preservation and degree of packing required (see 5.1 and 5.2).
f. When strapping shall be finish A (see 5.2.1).

6.3 First article ram. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional material, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each ram.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary, including any special tools, and instructions regarding shipment of rams.

Custodians:
Army - ME
Air Force - 99

Review activity:
Air Force - 84
DLA - CS

Preparing activity:
Army - ME

Project 4320-0296

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-52411B	Rams, Hydraulic, 10,000 PSI

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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